

GVEC: A GENERATIVE VIETNAMESE CHATBOT FOR ECONOMY, USING VIETNAMESE ECONOMY INFORMATION DATABASE FROM VNECONOMY COMMUNITY

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Abstract— Currently, the problem of automatic question answering (Q/A) with applications in chatbot services has become essential, addressing various issues in daily life. There are many approaches to solving this problem, but each presents its own challenges. The generative AI approach has recently gained widespread attention due to its ability to provide smooth, human-like responses. However, it has the drawback of producing coherent-sounding but inaccurate or fabricated content, known as "hallucinations". In this study, we aim to find a solution for the question-answering task in the context of the Vietnamese economy. We propose the Generative Vietnamese Economy Chatbot (GVEC), based on the Vietnam Economy Information Database (VEID) retrieved from VnEconomy systems. Our proposition is to apply Retrieval-Augmented Generation (RAG) to various LLM systems and test them on our specially designed benchmark, called the Vietnamese Numeric Economy Information Question/Answer Datasets (VNEIQAD), generated by our Question/Answers Generating with Numerical Information (QAGwNI) algorithm. This will help better evaluate the solution in the economic context.

Keywords— automatic question answering, chatbot, Vietnamese language, generative AI

I. INTRODUCTION

A chatbot, originally known as a chatterbot, is a software application or web interface designed to imitate human conversation through text or voice interactions. Modern chatbots are typically online and utilize artificial intelligence systems capable of engaging in natural language conversations with users, mimicking human conversational behavior. While more advanced chatbots often incorporate deep learning and natural language processing techniques, simpler versions have been in existence for many years. One significant application area for chatbots has been in customer service and support, where virtual assistants are commonly used. Companies across various industries have started leveraging the latest AI technologies to develop more advanced solutions in these areas.

Since late 2022, there has been widespread interest in the field, largely due to the popularity of OpenAI's ChatGPT, along with alternatives like Microsoft's Copilot and Google's Gemini. These examples highlight a recent trend of building such products on large foundational language models, such as GPT-4 [4] or the Gemini [9, 10, 11] language model which are then fine-tuned for specific tasks, such as simulating human

conversation in the case of chatbots. Furthermore, chatbots can be tailored or customized for specific situations or domains. As chatbots generate responses based on predictions rather than understanding the meaning behind them, they may produce coherent-sounding but inaccurate or fabricated content, known as "hallucinations". With the increasing adoption of chatbots for content generation, there are concerns about the potential impact on the proliferation and consumption of misleading information. This issue with generative AI systems is particularly severe in fields that demand high accuracy, and there is a significant error rate when using private datasets that are additionally trained into older generative models. Another problem for generative AI is the proof of true, or simply, the reference. Therefore, for these specific requirements, using generative AI systems hardly showcases their advantages. After an initial period of great excitement following the advent of generative AI systems like ChatGPT, their drawbacks have gradually become apparent. Although the smoothness and human-like quality of responses provided by generative AI systems are very appealing, their poor accuracy makes them less suitable for chatbot systems that require precision and source citation, especially when dealing with private information databases. Consequently, the combination of generative AI systems and information retrieval chatbot systems has begun to be explored recently.

Therefore, the demand for a chatbot system that can provide accurate information with clear source citations from learned databases, while also delivering smooth and human-like responses, is of great interest. Our system is designed to achieve this goal. In this research, our primary contribution is the development of a chatbot system specifically to address Vietnamese economic issues. This includes testing various architectures to select the optimal large language model using our economic benchmark, VNEIQAD, for the design of the Generative Vietnamese Economy Chatbots (GVEC) to provide correct answers with demonstrable authenticity.

II. VIETNAMESE LARGE LANGUAGE MODELS FOR CHATBOT

Today's chatbots are predominantly internet-based and utilize artificial intelligence to conduct conversations that resemble human interaction. Advanced chatbots often employ deep learning and natural language processing (NLP) to enhance their conversational capabilities, though simpler

versions have existed for quite some time. Various industries are increasingly adopting the latest AI advancements to create more sophisticated solutions in these fields. Currently, there are several popular approaches for chatbots such as the rule-based approach, the information retrieval-based approach, and the generative-based approach.

Rule-based approach: The rule-based approach uses heuristic rules to search for vocabulary and semantics in the questions and documents [12, 14]. This is considered the earliest Q&A approach. Q&A systems based on this approach follow a set of predefined rules or sets of instructions to generate answers to user input questions. They use a series of conditional statements to check for keywords or phrases in the user's input question and provide corresponding answers based on these conditions. The rule-based approach can be effective in certain cases but has some limitations, such as: (a) Limited natural language understanding, cause since these Q&A systems are built on predefined rules, they struggle with complex questions that do not match predefined patterns [15]; (b) Lack of context, because these systems cannot understand the context of the conversation, consequently they cannot grasp the user's intent beyond the predefined rules [12]; (c) No learning or adaptation, as these Q&A systems cannot learn or adapt, they cannot improve their responses over time, even if the responses become outdated or ineffective [13].

Information retrieval-based approach: The rule-based approach requires a structured set of predefined rules. To achieve this, they need resources to create these structured rules. The information retrieval-based approach does not require predefined rules; systems based on this approach only need unstructured documents and automatically extract information from them. Nowadays, obtaining unstructured information such as text documents and HTML pages is simple and not too complex. From these unstructured documents, collections of contexts, questions, and answers are generated semi-automatically. The information retrieval-based Q&A system is built on two main processes: document retrieval and answer extraction [16, 17, 18]. During document retrieval, the system must select documents from a large dataset relevant to the question. The selected documents are processed by the answer extraction process to return the most suitable answer. Since the conversational datasets are predefined, information retrieval methods do not have grammatical errors. However, they cannot handle cases that were not pre-trained [19].

Generative-based approach: Unlike the previous two approaches, the generative-based approach does not rely on predefined answers but generates entirely new responses. It uses large language models (LLMs) trained on a vast corpora with billions of words, phrases, and sentences [20, 21]. This approach combines deep learning algorithms and natural language processing techniques to understand the complexities of language and generate more human-like responses. Additionally, this approach can learn and improve its Q&A performance over time. Generative models are smarter than the previous two models and can reference prior entities. However, models following this approach are quite complex and often make grammatical errors with long questions. Moreover, they require a very large training dataset.

With the advent and success of ChatGPT, the generative-based approach has nearly triumphed and made a significant impact on the scientific community. However, after a few years, it has revealed weaknesses such as the phenomenon of

coherent-sounding but inaccurate or fabricated content, known as "hallucinations" [22]. Recently, trends have emerged that combine older approaches such as the rule-based approach and the information retrieval-based approach with the generative-based approach. We will discuss this generative-based approach in more detail in the following sections.

We explore several large language models that support the Vietnamese language: VinaLLaMA-7B-chat [1], Vistral-7B-chat [3], Openchat-3.5-7B [2], and GPT-3.5-turbo (also known as ChatGPT) as described by Brown et al. [4].

The VinaLLaMA [1] and Vistral [3] models result from the supervised fine-tuning (SFT) process [5] based on open-source models such as LLaMA-2 and Mistral, using Vietnamese text corpora. The Openchat-3.5 model developed by Wang et al. [2] is a fine-tuned version of LLaMA-2 using Reinforcement Learning from Human Feedback (RLHF) as described by Ouyang et al. [7] and Rafailov et al. [8]. Specifically, they introduced Conditional-RLHF, utilizing a multi-domain text corpus of mixed quality in English. They demonstrated that their model is the first 7B parameter model to achieve results comparable to ChatGPT. These open-source models are converted into the Q5_K GGUF format to be run locally on a robot, though this conversion may reduce performance. In addition to open-source models, we also considered GPT-3.5-turbo, which is accessed through the application programming interfaces (APIs) and OpenAI.

The issue of hallucinations, where language models generate text that appears coherent but is factually inaccurate, has become significant with the advent of those large models. In business contexts, many companies possess private documents, such as technical manuals and user guides. Using LLMs in these cases often leads to inaccuracies because they lack the context provided by private data. This challenge encourages the development of Retrieval-Augmented Generation (RAG). In fact, RAG supplements LLMs with reliable external information, providing sufficient context to generate more accurate responses. Additionally, RAG enhances the auditability of responses by citing sources, which allows for better verification and builds trust in the provided information.

Another possible option is to fine-tune LLMs with a factuality-aware alignment [23, 24, 25]. However, this method faces challenges due to frequent updates in data and the absence of clear traceability back to the original data sources.

In general, RAG involves indexing each paragraph in a document and, upon receiving a query, retrieving the most relevant paragraphs to be fed into a LLM. By providing selective paragraphs rather than the entire document, this approach prevents information overload and significantly improves response quality.

With RAG, large language models can leverage knowledge and information from external sources rather than relying solely on the data encoded in their model weights. This approach involves enhancing the information retrieval tool (retriever) to search for relevant external knowledge, such as databases, books, articles, and websites. By incorporating these external sources, RAG enriches the knowledge base available to LLMs.

The retriever tool in RAG can vary based on the semantic needs and can include:

Vector Database: This typically involves embedding query sentences using models like BERT (based on Transformers) [31, 32] to create dense vector embeddings. Additionally, traditional methods like TF-IDF [33] can be employed for sparse embeddings. The search is then conducted based on semantic similarity or term frequency.

Graph Database: This constructs a knowledge base from relationships between entities extracted from the text. While this approach ensures precise knowledge retrieval, it requires accurate queries, which may limit its applicability in some scenarios [34, 35].

Regular SQL Database: This provides storage and retrieval of structured data but may lack semantic flexibility. In the context of RAG, a regular SQL database plays a crucial role in managing structured data for efficient retrieval and integration with language models [36, 37].

In this study, we propose to validate the combinations of the RAG architecture (using a vector database) and the LLMs proposed above, using a proof-of-truth test dataset focused on economic information for question-answering. Other retrieval methods will be explored in our future work.

III. OUR PROPOSITIONS

A. Vietnamese Economy Database (VED) and Vietnamese Economy Q/A Pair Dataset (VEQAP)

Up to now, no official Vietnamese economy benchmark has been proposed to test the accuracy of the model. Therefore, we have created our own Vietnamese economy benchmark to conduct the accuracy tests. We designed a system that automatically generates queries and answers from the source data from the VnEconomy community, the largest Vietnamese economy information until now, called Vietnamese Economy Information Database (VEID). Currently, in Vietnam, we can list several sources of Vietnamese economy information as follows:

- VnEconomy, by Vietnam Economic Science Association (in Vietnamese: Hội Khoa học Kinh tế Việt Nam): <https://vneconomy.vn>
- CafeF by VCCorp. : <https://cafef.vn>
- Business Section in Tuoitre newspapers: <https://tuoitre.vn/kinh-doanh.htm>
- Economy Section in Thanh Niên newspapers: <https://thanhnien.vn/kinh-te.htm>

We employed a Python script to retrieve all economic information data from the publishing website of VnEconomy, the largest source of Vietnamese economy information and analysis. The retrieved data was saved as text files (.txt, .pdf, .html) to form the original database. This original database was then structured into hierarchical files. Subsequently, the data underwent reprocessing through sentence segmentation and tokenization based on a dictionary. Finally, the data was stored in a hierarchical structure in *json* format, creating the Vietnamese Economy Information Database (VEID). The VEID includes nine main categories: enterprise (in Vietnamese: doanh nghiệp), digital economy (in Vietnamese: kinh tế số), finance (in Vietnamese: tài chính), world event (in Vietnamese: thế giới), market (in Vietnamese: thị trường), real estate (in Vietnamese: bất động sản), livelihood (in Vietnamese: dân sinh), investment (in Vietnamese: đầu tư),

and spotlight (in Vietnamese: tiêu điểm). The database VEID contains 27,682 documents, totaling 30,186,806 words.

To verify the accuracy of chatbot responses, independent testing on a specialized benchmark is required. Currently, no such benchmark exists. Therefore, we aim to develop an independent benchmark to evaluate the accuracy of the optimal model. From the VEID, we generated question-answer pairs using various methods for different purposes. In this paper, we apply the Numerical Question/Answers Method instead of traditional methods. Given that chatbot responses are human-like, evaluating the quality of answers to general questions is challenging. Therefore, to ensure accurate assessment, we will focus on responses that include precise numerical answers. A response is considered correct if it contains the exact number asked for in the question and is accompanied by proof of truth, i.e., a reference to the source of the answer. If either the numerical answer or the reference is incorrect, the response is deemed wrong. Thus, the chatbot must provide the exact numerical answer and correctly reference the source of the information for the response to be considered accurate.

We developed a system for generating questions and answers (benchmark), where the answers are related to economic provisions. We created two benchmark sets: one consisting of 100 random questions from the database (VNEIQAD-R100) and another encompassing all questions in the database (VNEIQAD-ALL). The tool we developed generates questions from each sentence containing numerical information in the database. We refer to it as the Vietnamese Numeric Economy Information Question/Answer Datasets (VNEIQAD), with subsets VNEIQAD-R100 and VNEIQAD-ALL. The algorithm, named Question/Answers Generating with Numerical Information (QAGwNI), is implemented in Python. The flow chart for our QAGwNI system is provided below, see *Fig.1*.

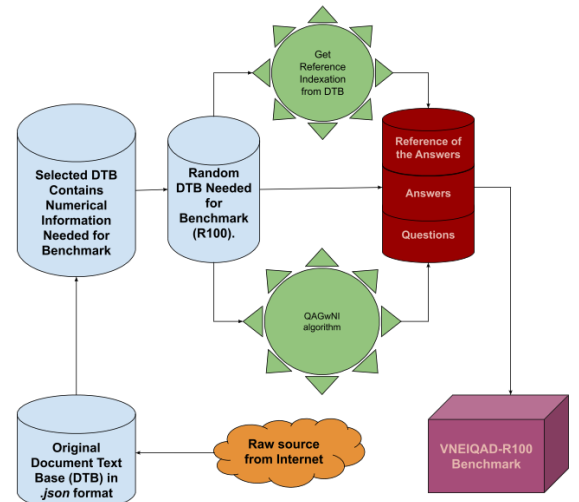


Fig. 1. Our Design for Automatically Generating QAGwNI Benchmarks

B. Generative Vietnamese Economy Chatbot (GVEC)

In this section, we explain our RAG design and how the proposed LLMs (Table I) are integrated into it. This RAG design is used in our Vietnamese economy chatbot. All collected data is pre-processed and splitted into paragraphs. Those paragraphs are vectorized and stored in the vector database. For each query, the retriever returns top-K paragraphs relevant to the query. The query and those

paragraphs are merged to form an input prompt of a LLM. This LLM generates an answer based on this prompt. See Fig.2.

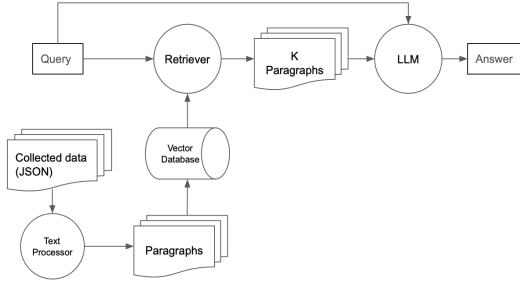


Fig. 2. Our RAG design.

From the raw data in website formats, the systems will crawl and reformat each article to various formats: *.txt*, *.pdf*, and *.html*, and also remove noise data (example: advertisement) to constitute the original data. This original data undergoes language processing (segmenting paragraphs, splitting sentences, tokenizing, creating hierarchical structures, removing junk data, etc.) to form the source database in *json* format [27, 28, 29, 30].

The source database is further processed into a Vector Database for RAG tasks, which involves five steps:

1. Create a vector database: first, convert all knowledge data into vectors and store them in a vector database. In detail, each vector represents a text paragraph in our json files.
2. Send a query to the chatbot: a user provides a natural language query to search for an answer or to complete the query. This query is later converted into a vector for the information retrieval.
3. Retrieve information from the vector database: The retrieval mechanism scans all vectors in the database to identify a fixed K number of knowledge segments (i.e., text paragraphs) that are the most related to the user's query, based on cosine distance. These paragraphs are then fed into the LLM to enhance context for generating an answer. In our experiment, $K = 4$.
4. Create a prompt: the paragraphs retrieved from the database are combined with the user's initial query to form a prompt.
5. Generate an answer: the prompt, supplemented with additional context, is passed through the LLM to generate the final response according to the enriched context.

The predicted responses of these LLMs are compared with the ground truth in the benchmark dataset to determine the optimal model. The optimal model is then implemented in the final product's system.

TABLE I. LARGE LANGUAGE MODELS USED IN OUR WORK

Model	# Params	Context length (# tokens)	Base model	Access
VinaLLaMA-7B-chat	7B	4 096	LLaMA-2	Q5_K GGUFF
Vistral-7b-chat	7B	32 768	Mistral-7B-v0.1	Q5_K GGUFF
Openchat-3.5-7B	7B	8 192	LLaMA-2	Q5_K GGUFF
Gpt-3.5-turbo	Undisclosed	16 385	From scratch	API

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Openchat-3.5-7B	7B	8 192	LLaMA-2	Q5_K GGUFF
Gpt-3.5-turbo	Undisclosed	16 385	From scratch	API

IV. EXPERIMENTS

In our experiment, we will compare four different LLMs (ChatGPT, OpenChat, VinaLLaMA, and Vistral) in a RAG architecture with our proposed data VEID. Recall that ChatGPT is integrated through API and other LLMs are locally run on our system (see Table I).

To evaluate their comprehensibility of the VEID, we use the numeric testset VNEIQAD-R100 consisting 100 pairs question/answer. Those pairs are randomly generated by the QAGwNI algorithm proposed in the previous section. In detail, the answers contain the keywords i.e., a number, a percentage or a datetime. An answer is considered accurate if it includes all of those keywords. Conversely, it is deemed inaccurate if it contains only a part of the keywords.

We use HE-EM (Human Evaluation Exact Match) to measure the capacity of extracting those keywords from the database and constructing the human-like answers. We manually evaluate the correctness of an answer based on its ground truth. This is a relatively simple measure. For a Q&A pair, if the keywords in the predicted answer exactly match the keywords in the reference answer, then $EM = 1$; otherwise, $EM = 0$. Recall that our ground truth is an answer with precise numbers. Therefore, the score of an answer is equal to 1 if the chatbot provides the correct numbers in the answer. We also use BLEU score [26] to observe the sentence similarity based on n-grams. BLEU = 1 indicates a perfect match between the generated answer and ground truth.

TABLE II. THE RESULT OF 4 LARGE LANGUAGE MODELS USED IN OUR EXPERIMENTS

Model	BLEU	HE-EM
VinaLLaMA-7B-chat	0.16	0.35
Vistral-7b-chat	0.58	0.74
Openchat-3.5-7B	0.10	0.55
Gpt-3.5-turbo	0.62	0.79

Based on the HE-EM results (Table II) and the nature of the experimented models (Table I), several observations about their correlation can be made. This VinaLLaMA model has the lowest EM score among the four models (< 0.5), indicating its weakest ability in providing exact matches to the ground truth answers. The lower EM score might be attributed to factors such as the model's architecture and the fine-tuned data. With a context length of 4 096 tokens, it might struggle to capture and retain relevant information for accurate responses, especially in cases where contextual understanding is crucial. Moreover, its fine-tuned data mainly includes a

collection of books, where only a small portion of books (7.5%) is about finance [3]. This could lead to a misunderstanding of the Vietnamese economy context in most questions of our benchmark.

A similar performance is obtained by OpenChat model (HE-EM = 0.55), which has the same base model LLaMA-2. Moreover, OpenChat and VinaLLaMA also obtain the smallest BLEU scores. It seems that the models based on LLaMA-2 meet difficulties in understanding our proposed database.

Vistral-7B-chat achieves the highest HE-EM score (0.74) among the open-source models, suggesting that it performs better in providing exact matches to the ground truth answers. With a larger context length of 32 768 tokens and based on the Mistral base model, it has a larger capacity to retain and utilize contextual information, leading to more accurate responses.

It is quite surprisingly that Gpt-3.5-turbo only help to improve the chatbot performance of 0.05 HE-EM score, compared with Vistral-7B-chat. Recall that Vistral-7B-chat only has 7B parameters, significantly fewer parameters than Gpt-3.5-turbo. Moreover, it is a quantized model in the Q5_K GGUFF format to be locally run on our system. This clearly indicates that a compressed open-source LLM can obtain a comparable result with the uncompressed LLM when it is used in a specific domain like the Vietnamese economy.

In general, the configuration of each model, including factors such as context length and base model, can impact its HE-EM score. Models with larger context lengths and potentially more sophisticated architectures or training strategies might have a better ability to understand and generate accurate responses, resulting in higher EM scores. Conversely, models with shorter context lengths or less sophisticated architectures might struggle to capture and retain relevant information, leading to lower HE-EM scores.

TABLE III. SAMPLE ANSWERS PRODUCED BY 4 MODELS + RAG WITH THE QUESTION AND ITS GROUND TRUTH (GOOD RESULT)

	Answers
Question	Tuy nhiên, có bao nhiêu % khảo sát cho rằng chi phí đầu tư cho an toàn thông tin chưa đến bao nhiêu % chi phí công nghệ thông tin của đơn vị? (En. However, how many % of surveys believe that investment costs for information security are less than how many % of the unit's information technology costs?)
Answers by ChatGpt + RAG	24% khảo sát cho rằng chi phí đầu tư cho an toàn thông tin chưa đến 5% chi phí công nghệ thông tin của đơn vị. (En. 24% of the survey said that investment costs for information security are less than 5% of the unit's information technology costs.)
Answers by OpenChat + RAG	24% khảo sát cho rằng chi phí đầu tư cho an toàn thông tin chưa đến 5% chi phí công nghệ thông tin của đơn vị. (En. 24% of the survey said that investment costs for information security are less than 5% of the unit's information technology costs.)
Answers by Vinallama + RAG	Câu trả lời là: 25% (En. The answer is: 25%)
Answers by Vistral + RAG	24% khảo sát cho rằng chi phí đầu tư cho an toàn thông tin chưa đến 5% chi phí công nghệ thông tin của đơn vị. (En. 24% of the survey said that investment costs for information security are less than 5% of the unit's information technology costs.)
Ground Truth	Tuy nhiên, có 24% khảo sát cho rằng chi phí đầu tư cho an toàn thông tin chưa đến 5% chi phí công nghệ thông tin của đơn vị. (En. However, 24% of surveys said that investment costs for information security are less than 5% of the unit's information technology costs.)

TABLE IV. SAMPLE ANSWERS PRODUCED BY 4 MODELS + RAG WITH THE QUESTION AND ITS GROUND TRUTH (BAD RESULT)

	Answers
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Question	Định giá khoảng bao nhiêu tỷ USD, công ty AI này hy vọng sẽ đảm bảo được nhiều nguồn tài trợ hơn vì đây có vẻ là thách thức lớn đối với OpenAI? (En. Valued at approximately how many billion USD, does this AI company hope to secure more funding as this seems to be a major challenge for OpenAI?)
Answers by ChatGpt + RAG	Theo các thông tin trong đoạn văn, không có thông tin cụ thể về việc định giá công ty AI nào đó bao nhiêu tỷ USD để đảm bảo được nhiều nguồn tài trợ hơn. Đoạn văn chỉ đề cập đến việc các công ty AI như OpenAI đang đối mặt với thách thức lớn từ các công ty khác như Google trong cuộc đua AI. (En. According to the information in the passage, there is no specific information about how many billions of dollars a certain AI company should be valued to secure more funding. The passage only mentions that AI companies like OpenAI are facing a big challenge from other companies like Google in the AI race.)
Answers by OpenChat + RAG	
Answers by Vinallama + RAG	là OpenAI không cung cấp con số chi tiết về số lượng người dùng của công ty AI này. (En. OpenAI does not provide detailed figures on the number of users of this AI company.)
Answers by Vistral + RAG	Google
Ground Truth	Định giá khoảng 18 tỷ USD, công ty AI này hy vọng sẽ đảm bảo được nhiều nguồn tài trợ hơn vì đây có vẻ là thách thức lớn đối với OpenAI. (En. Valued at around \$18 billion, the AI company hopes to secure more funding as it looks to be a major challenge for OpenAI.)

In the first example (Table III.), regarding the percentage of survey respondents indicating the investment cost for information security compared to the IT cost of the unit, VinaLLaMA-7B-chat and Vistral-7B-chat both produced similar responses, stating "24% of surveys indicate that the investment cost for information security is less than 5% of the IT cost of the unit". OpenChat-3.5-7B offered a slightly different response, stating "the answer is: 25%", which deviates from the ground truth. On the other hand, Gpt-3.5-turbo provided a more accurate response, matching the ground truth exactly.

In the second example (Table IV.), regarding the valuation of an AI company aiming to ensure more funding, ChatGpt + RAG and OpenChat + RAG both failed to provide relevant answers, leaving the responses blank and providing unrelated information, respectively. VinaLLaMA + RAG's response was incorrect, stating "OpenAI does not provide detailed information on the number of users of this AI company". Vistral + RAG's response was the most relevant, mentioning "Google", although it was not an exact match to the ground truth.

The accuracy of the responses varies among the models, with Gpt-3.5-turbo generally performing better in providing exact matches to the ground truth, followed by Vistral-7B-chat. VinaLLaMA-7B-chat and OpenChat-3.5-7B exhibited lower accuracy, with VinaLLaMA-7B-chat having the lowest HE-EM score. The differences in response accuracy can be partly attributed to the configurations of the models as indicated in the initial data table. Models with larger context lengths such as Gpt-3.5-turbo and Vistral-7B-chat, may have better capabilities to understand and generate accurate responses, leading to higher HE-EM scores. Conversely, models with smaller context lengths, such as VinaLLaMA-7B-chat and OpenChat-3.5-7B, may struggle to capture and retain relevant information, resulting in lower HE-EM scores. Additionally, differences in the underlying architectures and training data of the models can also influence their performance in providing accurate answers.

V. CONCLUSION

The challenge of automatic question answering (QA) for chatbot services has become increasingly important, addressing various everyday issues. Numerous approaches exist to tackle this problem, each with its own set of challenges. Recently, the generative AI approach has garnered significant attention due to its capability to produce natural, human-like responses. However, it has the limitation of generating coherent yet inaccurate or fabricated content, commonly referred to as "*hallucinations*". This study aims to address this issue by applying a solution to the QA task within the context of the Vietnamese economy. Our proposition is to apply Retrieval-Augmented Generation to various LLM systems for a question-answering system, called Generative Vietnamese Economy Chatbots (GVEC), implemented as a web service. This system is based on the Vietnam Economy Information Database (VEID) retrieved from VnEconomy systems.

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